

The Horseshoe Estimator for Sparse Signals

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Presentation

Observe a p -dimensional vector

$$\mathbf{y} \mid \boldsymbol{\theta} \sim N_p(\boldsymbol{\theta}, \sigma^2 \mathbf{I}).$$

Assume $\boldsymbol{\theta}$ is sparse: many entries near zero, a few large.

The estimator must simultaneously:

- Shrink noise to zero aggressively
- Leave true signals unshrunk

The Horseshoe Prior

$$\theta_i \mid \lambda_i \sim N(0, \lambda_i^2)$$

$$\lambda_i \mid \tau \sim C^+(0, \tau)$$

$$\tau \mid \sigma \sim C^+(0, \sigma)$$

$$\pi(\sigma^2) \propto 1/\sigma^2$$

where $C^+(0, a)$ is the half-Cauchy distribution on $(0, \infty)$ with scale a .

- Each θ_i has its own local scale λ_i
- Shared global scale τ controls overall sparsity
- No user-chosen hyperparameters (fully default)

The Shrinkage Coefficient κ_i

Reparametrize: let $\kappa_i = 1/(1 + \lambda_i^2) \in (0, 1)$.

Under a normal scale-mixture prior (for $\sigma^2 = \tau^2 = 1$):

$$\mathbb{E}(\theta_i \mid y_i) = \{1 - \mathbb{E}(\kappa_i \mid y_i)\} y_i.$$

So κ_i is the shrinkage weight:

- $\kappa_i \approx 0 \Rightarrow$ no shrinkage (signal)
- $\kappa_i \approx 1 \Rightarrow$ total shrinkage (noise)

Half-Cauchy on λ_i implies a $\text{Beta}(1/2, 1/2)$ prior on κ_i (U-shaped), concentrated near $\{0, 1\}$.
The prior a priori “votes” for either a signal or noise.

The Horseshoe Density

The univariate density has no closed form but is sharply bounded:

$$\frac{K}{2} \log\left(1 + \frac{4}{\theta^2}\right) < p(\theta) < K \log\left(1 + \frac{2}{\theta^2}\right), \quad K = \frac{1}{(2\pi^3)^{1/2}}.$$

Two crucial features:

- Pole at zero: $\lim_{\theta \rightarrow 0} p(\theta) = \infty \Rightarrow$ aggressive shrinkage of noise
- Cauchy-like tails: $p(\theta) \sim \theta^{-2} \Rightarrow$ robustness to large signals

Compare:

- Double-exponential: no pole, exponential tails (too light)
- Cauchy: heavy tails but no pole at zero
- Horseshoe: both

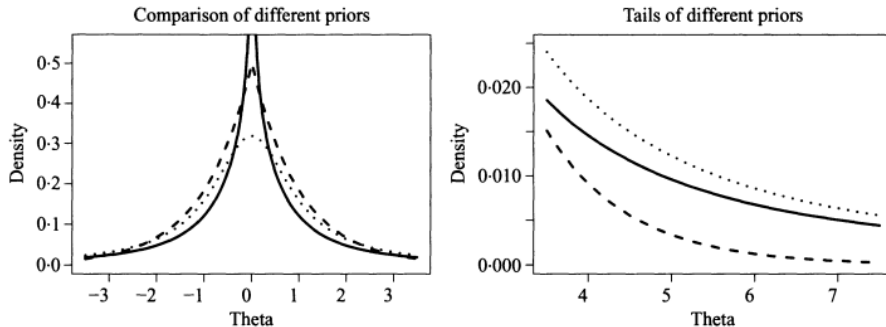


Fig. 1. Comparison of the horseshoe (solid), Cauchy (dotted) and double-exponential (dashed) densities.

Figure 1 plots the density in (1) with the standard double-exponential and standard Cauchy densities. The horseshoe prior has heavy, Cauchy-like tails decaying like θ^{-2} , along with a pole at $\theta = 0$. These key features allow the prior to perform well in handling sparse vectors.

Local Shrinkage Rules

Prior on θ_i	Density of λ_i	Density of κ_i
Double-exponential	$\lambda_i \exp(-\lambda_i^2/2)$	$\kappa_i^{-2} \exp\{-1/(2\kappa_i)\}$
Cauchy	$\lambda_i^{-2} \exp\{1/(2\lambda_i^2)\}$	$\kappa_i^{-1/2} (1 - \kappa_i)^{-3/2} \exp[\cdot]$
Strawderman-Berger	$\lambda_i (1 + \lambda_i^2)^{-3/2}$	$\kappa_i^{-1/2}$
Normal-Exp-Gamma	$\lambda_i (1 + \lambda_i^2)^{-(c+1)}$	κ_i^{c-1}
Normal-Jeffreys	λ_i^{-1}	$\kappa_i^{-1} (1 - \kappa_i)^{-1}$
Horseshoe	$(1 + \lambda_i^2)^{-1}$	$\kappa_i^{-1/2} (1 - \kappa_i)^{-1/2}$

The horseshoe is the only proper prior that is unbounded at both $\kappa = 0$ and $\kappa = 1$.

Beta(1/2, 1/2) is the unique “phase transition” point.

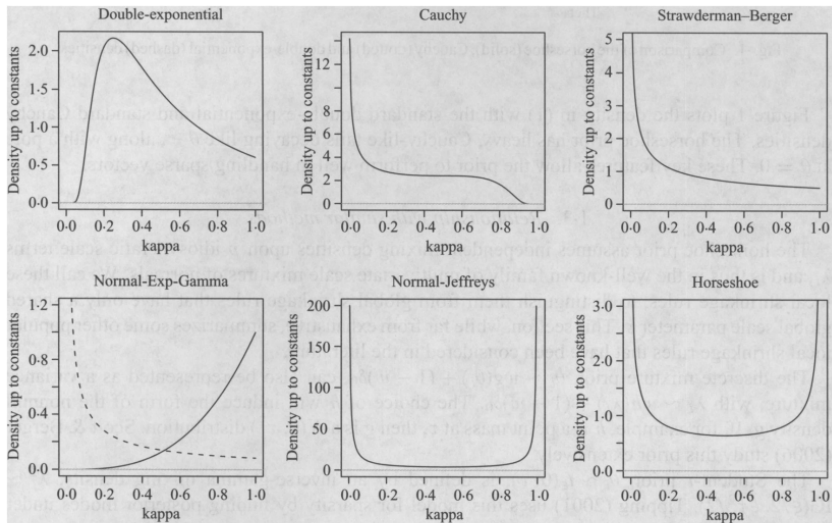


Fig. 2. The implied densities $p(\kappa_i)$ up to proportionality for six priors: the double exponential, Cauchy, Strawderman-Berger, normal-exponential-gamma, normal-Jeffreys and horseshoe. In the bottom-left panel for the normal-exponential prior, the solid line is for $c = 4$ and $d = 1$, while the dashed line is for $c = 1/4$ and $d = 1$.

Why not Lasso?

Under the double-exponential prior, κ_i density vanishes at 0:

$$p(\kappa_i) \propto \kappa_i^{-2} \exp\{-1/(2\kappa_i)\}, \quad p(\kappa_i) \rightarrow 0 \text{ as } \kappa_i \rightarrow 0.$$

Consequence: the Laplace prior cannot place enough mass on small κ_i (i.e., the no-shrinkage region).

Effect:

- Tail behavior: $|\mathbb{E}(\theta_i | y_i) - y_i| \approx \sqrt{2}/\tau$ for large $|y_i|$
- Small τ shrinks noise, but also overshrinks signals
- Large τ preserves signals, but leaves noise un-shrunk
- No single τ works well in sparse problems

Theorem 2: Posterior Mean Representation

Let $p(\theta)$ be any zero-mean scale mixture of normals with proper mixing density and finite marginal $m(y)$. Define

$$p^*(\lambda) = \lambda^2 p(\lambda), \quad p^*(\theta) = \int_{\mathbb{R}^+} p(\theta | \lambda) p^*(\lambda) d\lambda, \quad m^*(y) = \int_{\mathbb{R}} p(|y - \theta|) p^*(\theta) d\theta.$$

Theorem 2

$$\mathbb{E}(\theta | y) = \frac{m^*(y)}{m(y)} \frac{d}{dy} \log m^*(y) = \frac{1}{m(y)} \frac{d}{dy} m^*(y).$$

Under a normal likelihood, this reduces to

$$\mathbb{E}(\theta | y) = y + \frac{d}{dy} \log m(y).$$

Generalizes Masreliez (1975), Polson (1991) to unbounded priors.

Theorem 3: Bounded Influence

Theorem 3

Under the horseshoe prior with $\tau < \infty$:

$$|y - \mathbb{E}(\theta \mid y)| \leq b_\tau < \infty, \quad \lim_{|y| \rightarrow \infty} \frac{d}{dy} \log m(y) = 0.$$

Consequences:

- Large signals are left essentially unshrunk: $\mathbb{E}(\theta \mid y) \approx y$
- Risk bounded uniformly over all θ (Corollary 1): $\mathbb{E}\|\theta - \hat{\theta}^H\|^2 \leq p + pb_\tau^2$

Contrast: double-exponential has bounded but non-vanishing score $\Rightarrow$ persistent bias on large signals.

Horseshoe vs. Double-Exponential

Horseshoe score (as $|y| \rightarrow \infty$):

$$\frac{d}{dy} \log m(y) \rightarrow 0. \quad \text{Decreasing.}$$

Large signals: $\mathbb{E}(\theta \mid y) \rightarrow y$.

Double-exponential score (as $|y| \rightarrow \infty$):

$$\frac{d}{dy} \log m(y) \rightarrow -\sqrt{2}/\tau \neq 0. \quad \text{Bounded but persistent.}$$

Large signals: $\mathbb{E}(\theta \mid y) \approx y \pm b$, with bias $b = \sqrt{2}/\tau$.

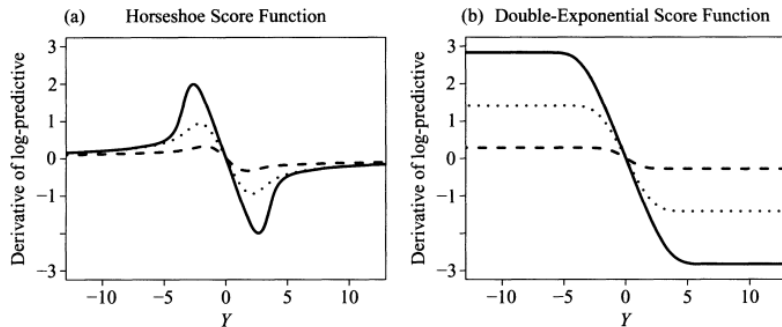


Fig. 3. A comparison of the score function for horseshoe and double-exponential priors for different values of the global scale parameter τ . (a) $\tau = 0.1$, solid line; $\tau = 1.0$, dotted line; $\tau = 10.0$, dashed line. (b) $\tau = 0.5$, solid line; $\tau = 1.0$, dotted line; $\tau = 4.0$, dashed line.

Joint Distribution: Sparsity via τ

Writing $\kappa_i = 1/(1 + \tau^2 \lambda_i^2)$, the joint density becomes

$$p(\mathbf{y}, \boldsymbol{\kappa}, \tau^2) \propto p(\tau^2) \tau^p \prod_{i=1}^p \frac{\exp(-\kappa_i y_i^2 / 2)}{(1 - \kappa_i)^{1/2}} \prod_{i=1}^p \frac{1}{\tau^2 \kappa_i + 1 - \kappa_i}.$$

Approximation when p is large (substitute $\bar{\kappa} = p^{-1} \sum \kappa_i$):

$$p(\tau^2 \mid \boldsymbol{\kappa}) \approx (\tau^2)^{-p/2} \exp\left\{-\frac{1}{\tau^2} \frac{p(1 - \bar{\kappa})}{\bar{\kappa}}\right\}.$$

τ is estimated by the average signal density:

- $\bar{\kappa} \approx 1$ (mostly noise) $\Rightarrow$ small τ a posteriori
- Connection to multiplicity correction (Scott & Berger, 2006)
- τ plays the role of inclusion probability w in discrete mixtures

Theorem 4: Kullback–Leibler Rates

Let p_{θ_0} be the true sampling model $y_j \sim N(\theta_0, \sigma^2)$, and R_n the Cesàro-average Kullback–Leibler risk of the posterior mean density estimator.

Theorem 4

① Under the horseshoe prior:

- If $\theta_0 = 0$: $R_n = O\{n^{-1}(\log n - b \log \log n)\}$ (super-efficient)
- If $\theta_0 \neq 0$: $R_n = O(n^{-1} \log n)$

② For any continuous prior $p(\theta)$ bounded above:

$$R_n = O(n^{-1} \log n), \quad \text{regardless of } \theta_0.$$

Super-efficiency at zero arises from the pole in the prior density: more prior mass in a neighborhood of $\theta_0 = 0$. No bounded prior shares this property.

Informal Thresholding Rule

Discrete mixture model gives posterior probability of inclusion w_i :

$$\hat{\theta}_i = w_i \mathbb{E}_g(\theta_i | y_i) \approx w_i y_i \quad (\text{heavy-tailed } g).$$

Horseshoe estimator:

$$\hat{\theta}_i = (1 - \hat{\kappa}_i) y_i.$$

Proposed rule

Declare θ_i a signal if $1 - \hat{\kappa}_i > 1/2$; otherwise call it noise.

Although $1 - \hat{\kappa}_i$ is not formally a posterior probability, empirically it tracks w_i closely (Table 2 in the paper).

Correspondence with the Gold Standard

Table 2: Posterior probabilities w_i (discrete mixture) vs. $1 - \hat{\kappa}_i$ (horseshoe), as percentages. 10 fixed signals, increasing noise.

Noise	Signal 1.0	Signal 2.0	Signal 3.0	Signal 4.0	Signal 5.0
25	20 (17)	27 (21)	35 (28)	44 (37)	54 (47)
100	10 (17)	27 (31)	69 (62)	95 (85)	100 (92)
500	2 (4)	6 (10)	35 (36)	91 (80)	100 (92)
2000	0 (1)	1 (2)	8 (9)	59 (54)	98 (90)
10000	0 (0)	0 (1)	3 (3)	32 (30)	96 (88)

- Numbers in parentheses are horseshoe shrinkage weights $1 - \hat{\kappa}_i$
- Double-exponential weights do not behave like w_i

Simulation Study: Realized Squared-Error Loss

Sparse model: $y_i \mid \theta_i \sim N(\theta_i, 1)$, $\theta_i \sim w t_\xi(0, 3) + (1 - w)\delta_0$. 500 datasets per cell.

Method	$w = 0.05$		$w = 0.2$		$w = 0.5$	
	$\xi = 2$	$\xi = 10$	$\xi = 2$	$\xi = 10$	$\xi = 2$	$\xi = 10$
MLE	250	248	249	251	252	251
Double-exponential	171	127	237	217	247	235
NEG ($c = 4, d = 3$)	121	121	134	134	186	183
Empirical-Bayes	32	38	111	129	417	442
NEG (best fixed)	33	39	96	98	179	178
Horseshoe	32	33	94	95	178	244

Horseshoe systematically dominates the double-exponential and default NEG, matches the best-tuned NEG, and beats Empirical-Bayes thresholding in signal-rich settings.

Vanguard Mutual-Fund Data

Setup: estimate Σ for $p = 59$ funds from $n = 86$ weekly returns. Model the Cholesky factor of Σ^{-1} via horseshoe regressions.

Out-of-sample missing-value imputation, risk relative to Bayesian Model Averaging:

	MLE	Lasso AND	Lasso OR	Horseshoe	BMA
Squared error	10.63	1.25	2.12	1.07	1.00
Absolute error	3.51	1.22	1.47	1.04	1.00

- Horseshoe nearly matches BMA at a tiny fraction of the cost
- Substantially outperforms both lasso variants
- Provides a near-BMA shrinkage rule via a single local prior

The horseshoe prior combines:

- Strong global shrinkage through τ
- Robust local adaptation through the λ_i
- Pole at zero for noise suppression $\Rightarrow$ super-efficient rate when $\theta_0 = 0$
- Cauchy-like tails for signal preservation $\Rightarrow$ redescending score, bounded risk
- No tuning parameters, fully default prior

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Thank you!

Questions?